🚀 LAWS OF MOTION

Introduction

The laws of motion, formulated by Isaac Newton, describe the relationship between the motion of an object and the forces acting on it. These laws provide the foundation for classical mechanics and help explain a wide range of physical phenomena. Understanding these laws is essential for analyzing motion and solving numerical problems in physics.

Force and Its Nature

Force is defined as an external influence that changes or tends to change the state of rest or uniform motion of an object. It is a vector quantity and has both magnitude and direction. Forces can be classified as contact forces, such as friction and normal force, and non-contact forces, such as gravitational force.

Force plays a crucial role in determining how an object moves and interacts with its surroundings.

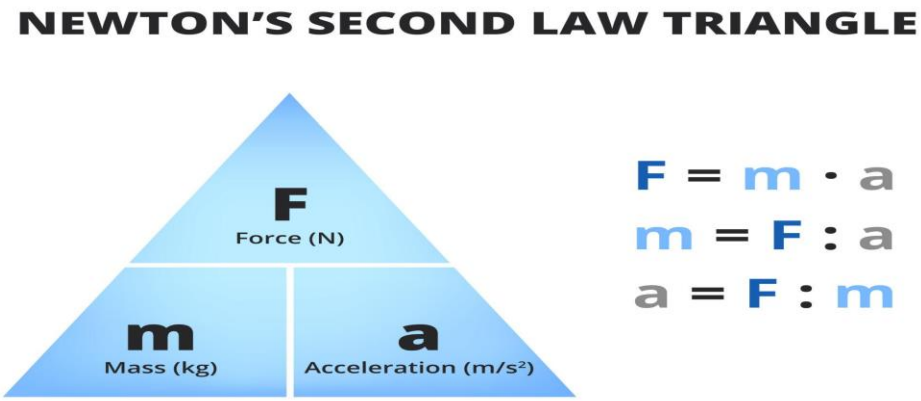
Newton's First Law of Motion

Newton's first law states that a body remains at rest or continues to move with uniform velocity in a straight line unless acted upon by an external force. This law introduces the concept of inertia, which is the tendency of an object to resist changes in its state of motion.

Objects with greater mass have greater inertia and require more force to change their motion. This law explains why a stationary object remains at rest and a moving object continues to move unless a force acts on it.

Newton’s Second Law of Motion

Newton’s second law provides a quantitative description of force. It states that the rate of change of momentum of an object is directly proportional to the applied force and occurs in the direction of the force.



Another way to look at Newton’s second law:

Define "linear momentum": $\mathbf{p} \equiv m\mathbf{v}$

The SI Unit of linear momentum is:

$$\text{kg} \cdot \text{m/s} = \text{s}\ (\text{kg} \cdot \text{m/s}^2) = N \cdot s$$

$$\boldsymbol{F} = \frac{d\mathbf{p}}{dt} = \boxed{\frac{d(m\boldsymbol{v})}{dt} = m\frac{d\boldsymbol{v}}{dt} + \boldsymbol{v}\frac{dm}{dt}} = m\boldsymbol{a} + \boldsymbol{v}\frac{dm}{dt}$$

For constant mass, $\boldsymbol{v}\frac{dm}{dt} = 0$, the relation becomes:

$$\boldsymbol{F} = m\boldsymbol{a}$$

This law is commonly expressed as (F = ma), where force is the product of mass and acceleration. It forms the basis for solving most numerical problems in mechanics.

Momentum

Momentum is defined as the product of mass and velocity of an object. It is a vector quantity and depends on both the mass and velocity of the object. The concept of momentum is important in understanding collisions and motion of objects.

The change in momentum of an object is directly related to the force applied and the time for which it acts.

Newton's Third Law of Motion

Newton's third law states that for every action, there is an equal and opposite reaction. This means that forces always occur in pairs, acting on different bodies.

This law explains various phenomena such as recoil of a gun, walking, and swimming. The action and reaction forces are equal in magnitude but opposite in direction and act on different objects.

Types of Forces

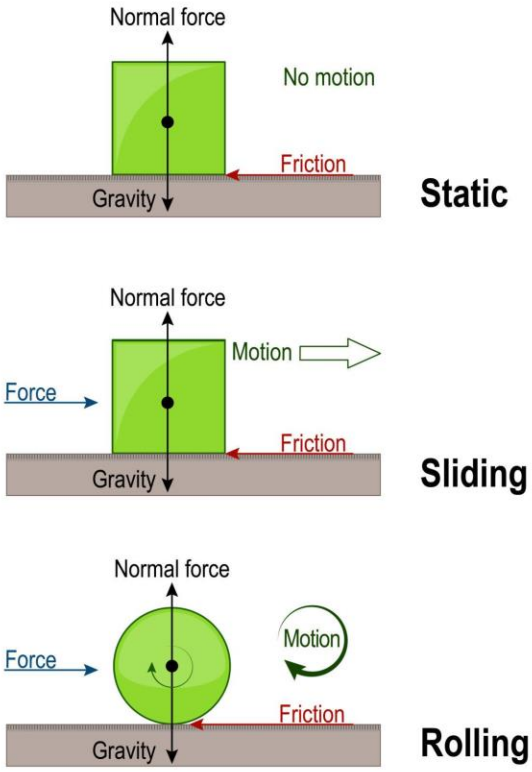
Different types of forces act on objects in various situations. The normal force is the force exerted by a surface on an object in contact with it. Friction is a force that opposes motion between two surfaces in contact. Tension is the force transmitted through a string or rope when it is pulled.

Understanding these forces is essential for solving problems involving multiple forces.

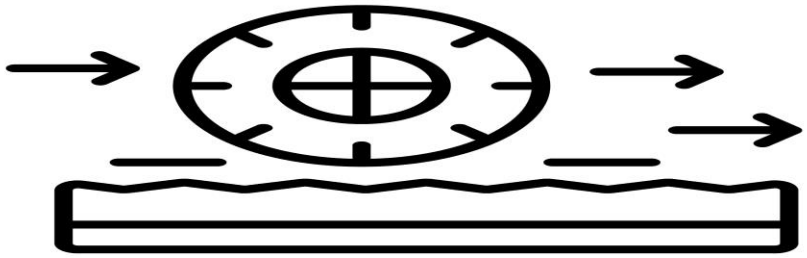
Friction

Friction is a resistive force that opposes the relative motion between two surfaces in contact. It arises due to irregularities on the surfaces and acts opposite to the direction of motion.

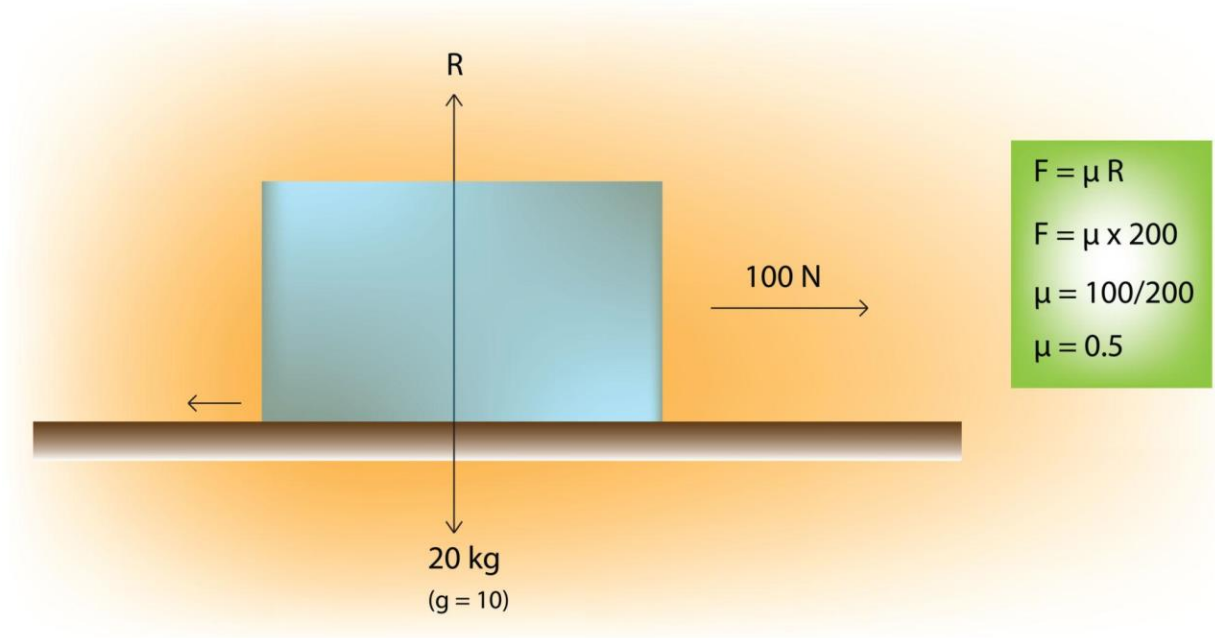
FRICTION

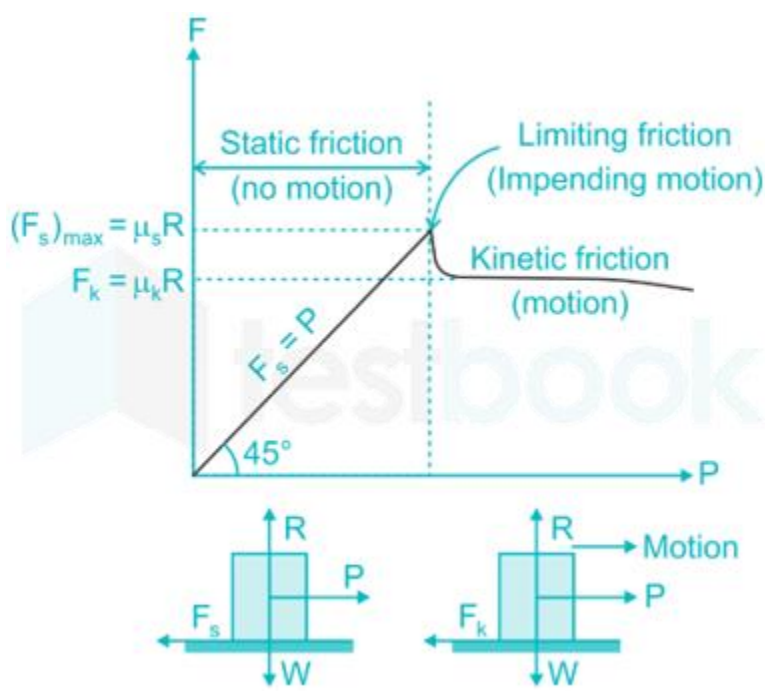


$$F = \mu N$$



Friction

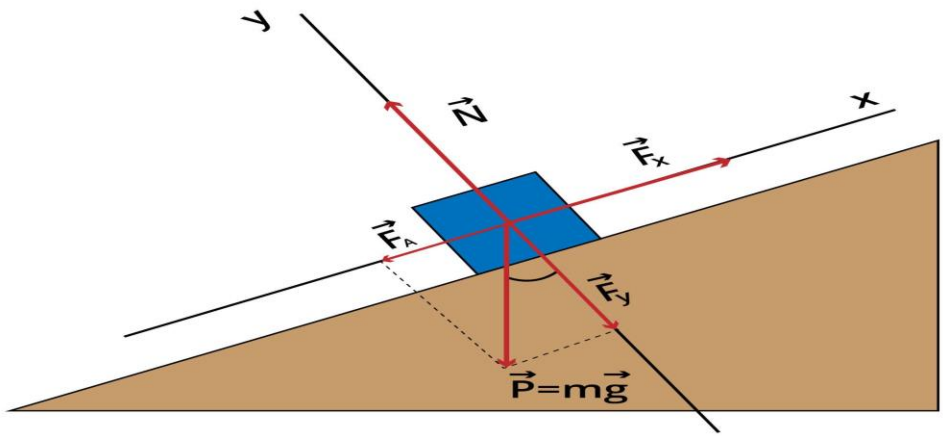




Friction depends on the nature of surfaces and the normal force acting between them. It plays an important role in everyday life and in mechanical systems.

Free Body Diagram

A free body diagram is a representation of all the forces acting on an object. It helps in analyzing motion by isolating the object and considering all forces acting on it.

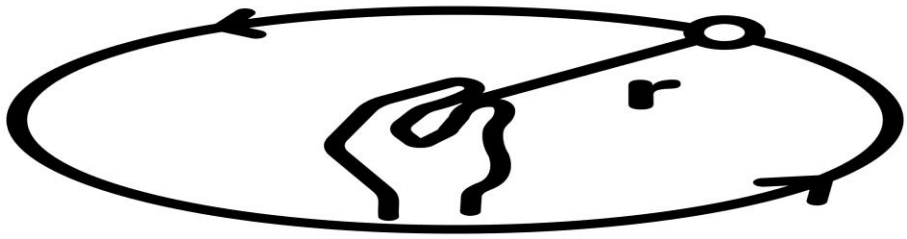


Free body diagrams are essential for solving problems involving multiple forces and are frequently used in NEET questions.

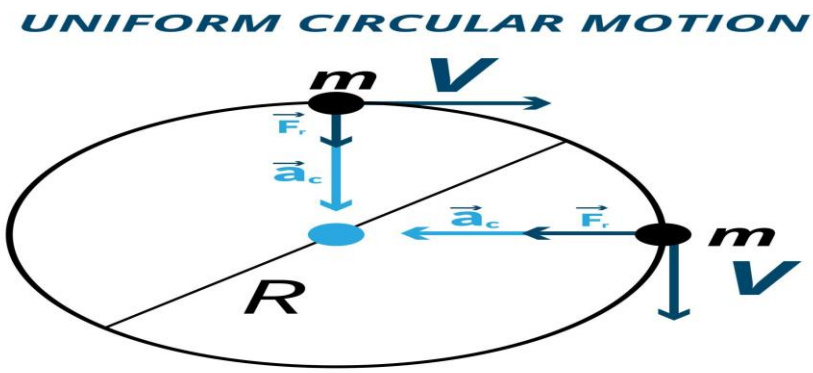
Circular Motion and Centripetal Force

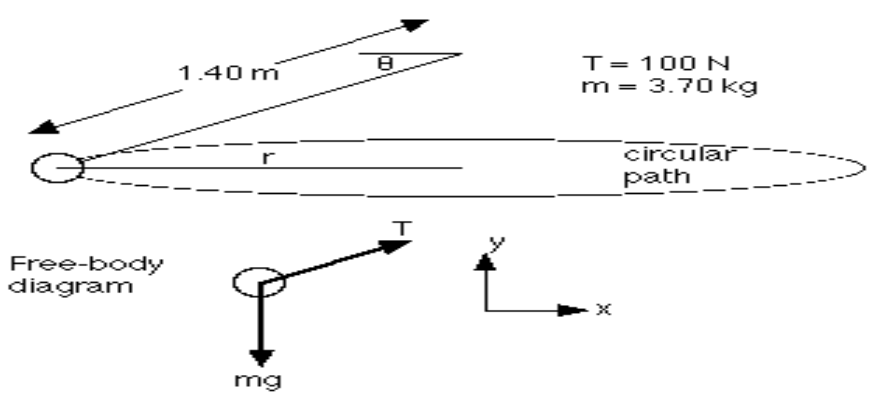
When an object moves in a circular path, it experiences a force directed toward the center of the circle, known as centripetal force. This force is responsible for keeping the object in circular motion.

$$F_c = \frac{mv^2}{r}$$



Centripetal Force





This concept is important in understanding motion of planets, vehicles on curved roads, and rotating systems.